



OPEN Debris variability and velocity estimation using Electrical Resistivity Tomography and DInSAR techniques of Durung Drung glacier, northwestern Himalaya

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This study integrates Electrical Resistivity Tomography (ERT), Synthetic Aperture Radar (SAR), and DGPS measurements to characterize the glacier dynamics, subsurface conditions, and moraine stability of the Durung Drung Glacier in the northwestern Himalaya. ERT profiles reaching depths of ~45 m resolves a distinct three-layer subsurface architecture composed of a low-resistivity, water-saturated surface layer, an intermediate zone of frozen sediments containing discontinuous ground-ice lenses, and a high-resistivity crystalline bedrock unit. The absence of massive, continuous ice within both the lateral and terminal moraines combined with the presence of isolated frozen pockets indicates melt-out of earlier ice-cored moraine complexes. This transition from ice-cored to ice-free but partially frozen moraine material indicates sharp thermal sensitivity, increased subsurface hydrological connectivity, and a growing susceptibility of moraine structures to internal weakening under continued climatic warming. Surface ice-flow velocities derived using Sentinel-1 A/B DInSAR, SAR offset tracking, and COSI-Corr feature tracking range from 120 to 138 m a⁻¹, capturing spatial variations in glacier motion across the ablation zone. Complementary DGPS surveys document a progressive snout retreat of 14.93 ± 0.5 m (2018–2019), 16.62 ± 0.5 m (2019–2020), and 21.43 ± 0.5 m (2020–2022), confirming rapid frontal recession. The frontal retreat is also driving the expansion of the moraine-dammed lake, as greater meltwater discharge and reduced ice support allow the lake to enlarge against a weakening terminal moraine. Subsurface debris characteristics, ice-flow dynamics, and hydrological pathways, providing a strong framework for evaluating moraine stability and glacial hazards in high-mountain Himalayan environments.

Keywords Durung Drung, NW Himalaya, Debris cover, Moraines, ERT, DInSAR

Over the past decade, there has been much discussion over the fate of the Himalayan glaciers, especially because of the impact of related glacial lake outburst floods (GLOFs) and their importance for future water supply scenario^{1,2}. However, this discussion has highlighted significant knowledge gaps and inconsistencies in forecasts regarding the future of Himalayan glaciers^{3,4}. The significant challenges in studying Himalayan glaciers are the limited availability of in-situ data, which often hinders their application and interpretation. Alterations to glacier dynamics i.e., mass balance, thickness and velocity can be directly linked to shifts in climate variables such

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as temperature and rainfall⁵. Monitoring the changing patterns in glaciers is crucial for understanding how rising sea levels are influenced by glacial meltwater, how various ice masses are responding to changes in the climate, and even how regional and local management of waters is affected⁶. Despite advancements in remote sensing techniques⁷ and numerical modelling methodologies⁸, the scarcity of on-site measurements remains a significant challenge. This is mostly due to the difficult terrain and the glaciers' high-altitude, isolated settings⁹. In the past, investigations of subsurface conditions in glacial environments often involved invasive methods such as coring and borehole surveys. However, these approaches have limitations as they can disrupt natural subsurface processes and are logistically challenging and expensive¹⁰. To overcome these drawbacks, electrical geophysical techniques have gained popularity in hydrological and glacier studies^{11,12}. These methods provide a cost-effective and minimally intrusive approach for investigating subsurface characteristics and processes over larger spatial scales but they are best used in conjunction with direct observations, such as borehole data, as geophysical techniques offer indirect interpretations and cannot fully replace ground-truth validation.

One widely used technique is Electrical Resistivity Tomography (ERT) which produces 2-D or 3-D images of the sub-surface's bulk electrical resistivity¹³. The bulk resistivity is inversely related to the bulk electrical conductivity. The resistivity of geological materials is primarily influenced by variables i.e., porosity, saturation and electrical conductivity of pore fluids^{13,14}. ERT has been extensively employed in hydrological and permafrost research to identify and characterize the ice and frozen ground. It is also effective in detecting variations in lithological and hydrological features in the subsurface. By using ERT, researchers can complement traditional access-hole surveys with a non-invasive and cost-effective approach. This allows for a better comprehension of subsurface hydrological features and processes on a larger scale, beyond individual points, while also increasing spatial coverage. ERT has proven valuable in studying glaciers and their hydrological systems, facilitating the detection of subsurface changes, delineation of hydrological features, and characterization of the distribution of ice and frozen ground¹¹.

In response to climate change-induced glacier retreat, moraine-dammed glacial lakes are fast forming throughout the Himalayas^{15,16}. These lakes can pose significant risks, as breaches in moraine dams may lead to catastrophic flooding events^{17,18}. Moreover, their presence can amplify glacial melt rates and accelerate long-term glacier retreat¹⁹. Despite their importance, there is limited understanding of the underlying features and inner dynamics of moraine dams, that play a crucial role in influencing their stability and evolution. Processes like seepage erosion and the breakdown of ice-cored moraine dams are responsible for the deterioration of terminal and lateral moraine structures²⁰.

The periodic mass balance, length, areal extension (melting) and volume (reducing) of glaciers can all be indicators of continuous climate change^{21–23}. However, gathering field data for the majority of Himalayan glaciers parameters, especially mass balance is limited by the distant location of the glaciers. In these situations, glaciers' responses to climate change are typically interpreted as changes in their overall length and extent²⁴. For the purpose of mapping and tracking glaciers, particularly their retreat, multispectral and multitemporal optical satellite data is widely used^{25,26}. As a result, field-based glacier tracking and mapping can be most effectively utilized in conjunction with remote sensing data. The number of debris-covered glaciers can significantly alter the response of these measurements, therefore it's important to carefully understand the signals they provide. In addition, glaciers' responses must be taken into account when analysing how their variations relate to change in climate parameters²⁷. It is difficult to estimate any previous shifts in the glaciers area because archives of hydro-meteorological information are nearly non-existent. Although it has been demonstrated that gridded climate datasets can be used for glacio-hydrological analyses²⁸, bias adjustment is necessary for these datasets. Glacier retreat is facilitated by the widespread increasing temperatures over the western Himalayas, which cause many proglacial lakes to originate and those already present trends to enlarge in size²⁹. Using field and remote sensing data, some researchers tried to investigate variations in glaciers taking into account different criteria as glacier dimensions, length, snout elevation, and glacier slope^{30–33}. This research demonstrated that the smaller, steeper glaciers had melted more quickly than the gradual, larger ones³⁴. Frontal, areal, and volumetric studies³⁵ are among those that concentrate on the influence of debris accumulation on glacier dynamics³⁶. These studies showed that pristine glaciers had higher retreat rates and areal reduction, whereas debris-covered glaciers showed fewer frontal modifications and larger areal decline.

Glacial lake outburst floods (GLOFs) are characterized by process cascades, which can result in high floods, debris floods, or debris flows when rock slope collapses, ice avalanches, or mass movements from moraines affect glacial lakes and create displacement waves that overturn and smash the dam³⁷. Lakes that have been blocked by glaciers and moraine can produce outburst floods, which are extremely mobile water and sediment combinations that can move tens of kilometers to over 100 km at speeds more than ten kilometers per hour. Their abrupt onset, massive discharge, long runoff distance, and propensity to flow along preexisting river courses where individuals and assets are concentrated make them a major threat³⁸. With regard to sediment entrainment and erosion, these events are extremely dynamic phenomena, and their volume and peak discharge can rise by a number of three or more in comparison to their original values³⁹.

The Western Ladakh (WL) glaciers had a high debris cover than Eastern Ladakh (EL) glaciers, which is often thought to decrease melt, but the warming had a greater effect on their low-angle debris-covered tongues. Furthermore, it was shown that surface slope played a significant role in mass balance variability in both areas, accounting for 30–42% of the observed variance. The main conclusions show that during the past 20 years, WL glaciers have lost more mass than EL glaciers⁴⁰. According to research on globally dispersed debris thickness, the worldwide median debris thickness is around 0.15 ± 0.06 m, which has been confirmed by data from 22 glaciers. By taking debris into account, sub-debris melt is reduced by an average of 37%, which can have an annual impact on the regional mass balance of up to $0.40 \text{ m w.e. a}^{-1}$ ⁴¹.

Minor surface variations measured in centimetres, could be found by paring the phase shift among two backscattered SAR reflections of the same area obtained at somewhat distinct viewing angles. It has been

demonstrated that the Differential Interferometry SAR (DInSAR) method is effective in identifying surface shifts brought by mass shifts in alpine environment. According to some research on the deformation of rock glaciers, the method may also work well for fairly tiny things (small movements in the earth surface) in hilly areas⁴². The DInSAR approach has been extensively employed in several locations to detect and measure the motion of landforms in the periglacial setting⁴³. One of the technique's primary capacities is that it offers reliable data on the occurrence of mobility. Furthermore, the DInSAR method makes it possible to identify the gradual surface motions that distinguish dynamic from dormant rock glaciers⁴⁴. Among the constraints of optical data, the DInSAR method can realistically and effectively resolve certain of them. It can function in the existence of clouds and snow, revealing insight on the landform's movement. Although continual temporal assessments are provided by conventional in situ techniques for surface dynamics (such as inclinometers and global navigation satellite system measurements), its spatial reach is still constrained⁴⁵. While element monitoring and digital elevation model (DEM) differentiation techniques offer spatially consistent knowledge, they frequently necessitate time-consuming, recurrent unmanned aerial vehicle collections on the scene, particularly when attempting to track dynamics on smaller time frames or during specific seasons⁴⁶. Due to these obstacles, space borne DInSAR, which depends on active synthetic aperture radar satellites, has emerged as a useful technique for obtaining glacier surface dynamics at an accessible cost, a wide geographic reach, and with significant spatiotemporal accuracy⁴⁷. Yet, the majority of DInSAR research concentrate on the enormous scale identification of moving rock glaciers, measurement of surface dynamics at the level of complete rock glacier sections⁴⁸. The aim of this study is to investigate the subsurface and surface dynamics of the moraine system in order to characterize the distribution and variability of debris materials, rock formations, and frozen sediments. By integrating geophysical and remote sensing techniques, the study seeks to understand the internal structure and evolution of lateral and terminal moraines, evaluate the influence of debris cover on glacier stability, and assess potential geomorphological hazards associated with moraine-dammed regions.

Study area and geological setup

The study focuses on the Durung Drung glacier, which is situated in the Zaskar basin of the Greater Himalayas in India (Fig. 1). With an average about 250 mm of precipitation per year, Zaskar is classified as a cold, arid, high-altitude area⁴⁹. Stretching 23.3 km, the Durung Drung glacier is a huge, clean ice glacier. It is 5260 m above

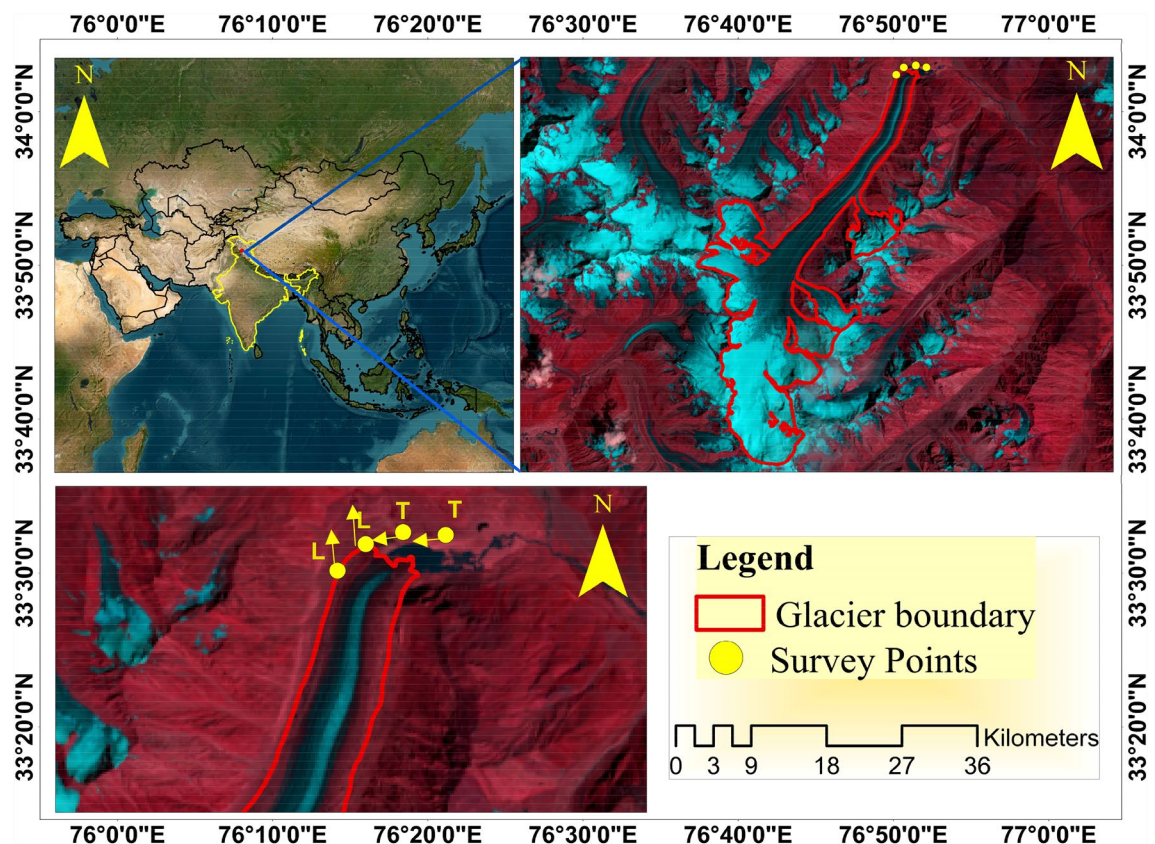


Fig. 1. An illustration showing the location of the study area, overlaid on a 2017 Landsat 8 Operational Land Imager (OLI) image, with a close-up view of the Durung Drung Glacier highlighting the Electrical Resistivity Tomography (ERT) field points (L = Lateral Moraine, T = Terminal Moraine; yellow arrow indicates the direction of ERT profile alignment). Landsat 8 imagery was obtained from the United States Geological Survey (USGS) Earth Explorer archive (<https://earthexplorer.usgs.gov/>) and is publicly available. Image processing and cartographic preparation were carried out using ArcGIS version 10.8.1 (<https://www.esri.com/en-us/home>).

sea level on average. This glacier provides meltwater to the Zaskar River, a tributary of the Indus, which is vital to the hydrology of the area⁵⁰. The Durung Drung Glacier is a valley-type glacier possesses mean slope of 60 degrees. The ablation zone is significantly narrower than the accumulation zone, which has a broad, gentle slope. The glacier's lateral and recessional moraines are indicative of its past extent and shifting behavior. The Durung Drung glacier is distinguished from other glaciers by its comparatively clean appearance and low debris cover. The glacier is a primary resource of fresh water for the region, feeding into the Stod River, which eventually merges into the Zaskar River.

During the winter, a heavy covering of snow covers nearly the whole valley. The wintertime temperature in the Zaskar Valley is about 2 °C. The summer season begins in May, and the temperature rises rather quickly. The summer season in Ladakh is somewhat longer than in other locations, lasting into August. In the winter, mid-latitude westerlies give moisture for the heavier snowfall that falls in the more western and northern highlands⁵¹.

Geological setup

The Durung Drung glacier has developed terminal and lateral moraines, as well as medial moraines. Notably, a loop-shaped terminal moraine is present. These moraines enclose inter-morainal lakes of various shapes and sizes. The junction of lateral moraines with terminal moraines results in the formation of box-shaped features. The southern portion of the Ladakh Himalaya is covered by the Zaskar basin, a huge mountainous area that lies between the Tethyan Himalaya and the Great Himalaya Range. The Zaskar fault (South Tibetan Detachment System), which divides the Tethyan Himalaya's meta-sediments from the Himalayan Crystalline Zone, is where the Durung Drung glacier is located. The Quaternary glacial deposits, boulders, scree and talus deposits, and frost-cracked rocks all have a significant impact on the geomorphology of the Greater Himalaya Range. These materials are then reworked and carried by mass movement, snowmelt, and glacial runoff processes⁴⁹.

During the late Pleistocene–Holocene, resurfacing along the Zaskar Fault caused displacement near the west (Pensi La), directing the Durung Drung Glacier northward and slightly northeast. The eastern portion of the glacier now shows an average dip of 25°. In the western sector, regional tectonics created a multi-stratified glacier mass. This layered structure can give the appearance of locally thicker ice, which may be misinterpreted as glacier advance in cross-sectional profiles or remote sensing images. Therefore, the apparent increase in glacier thickness reflects geological structures rather than actual glacier growth⁵², (Fig 2).

Datasets

Landsat's Multispectral Scanner (MSS), Thematic Mapper (TM), Enhanced Thematic Mapper Plus (ETM+), and Operational Land Imager (OLI) sensors have been used to collect multitemporal satellite data for a number of years, including 1976, 1980, 1990, 2000, 2010, 2015, 2020 and 2022. Landsat satellite sensors have been extensively employed for a number of purposes, such as studying glacier facies⁵³, monitoring glaciers⁵⁴, tracking snow line elevation⁵⁵, and analyzing glacier dynamics³. All of the Landsat sensors' detailed specifications are available at: <http://landsat.gsfc.nasa.gov/about/technical.html>.

DIInSAR, one of the latest and more precise remote sensing techniques for determining glacier movement, was used to create velocity maps of the Durung Drung Glacier. The two-pass interferometry technique was employed to create the interferograms, with the accuracy of the results depending significantly on the selection of suitable interferometric image pairs⁵⁶. For cryospheric studies in the Himalayan region, L-band SAR data is particularly advantageous due to its superior penetration capabilities through clouds and snow compared to C-band (e.g., Sentinel-1, Radarsat-2) and X-band (e.g., TerraSAR-X) datasets⁵⁶. This characteristic makes L-band data highly effective for analyzing glacier surfaces and subsurface layers that are less affected by short-term regional weather changes. While Sentinel-1 A/1B's C-band has demonstrated good coherence for certain glaciers (Mahagaonkar et al., 2019), especially with its shorter temporal resolution of 6–12 days, L-band data is preferred due to its deeper backscatter penetration and higher temporal stability, datasets like ALOS PALSAR-2, with a 14-day temporal baseline and TerraSAR-X (11 days), are commonly utilized for such analyses. For glacier



Fig. 2. Field photographs (a, b) during the resistivity survey using the Syscal Pro instrument.

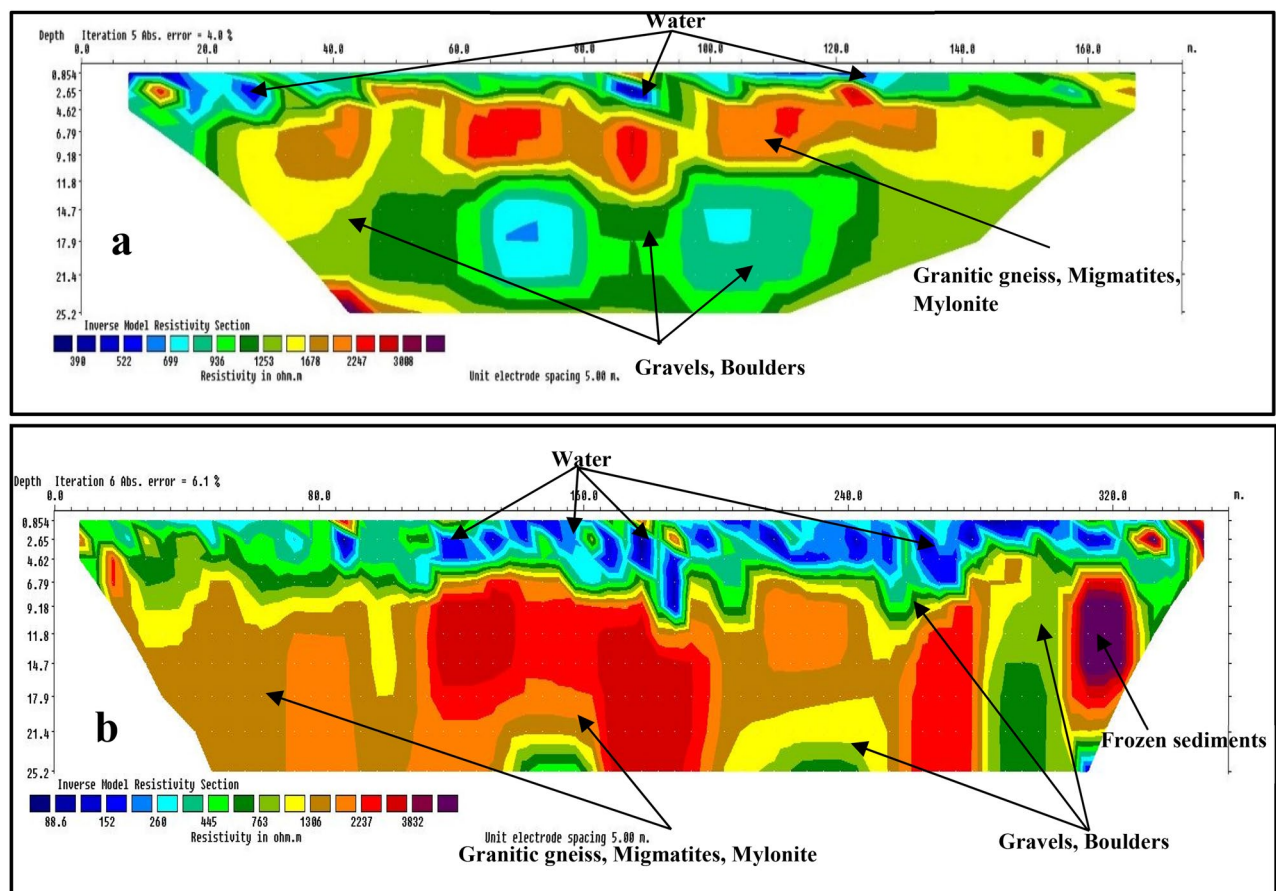


Fig. 3. Resistivity sections obtained after inversion of ERT lines (a) and (b) carried out along the Lateral Moraine of the Durung Drung glacier (Dipole-Dipole).

movement studies, SAR data with longer wavelengths and shorter temporal intervals are ideal needs⁵⁶. For this study, the Sentinel-1 A/B datasets were utilized, offering C-band SAR data (polarization VV or HH) with a wavelength of approximately 5.6 cm. Acquired in Interferometric width with a spatial resolution of 5*20 m, and a swath of 250 km, this dataset the SLC images with VV polarization acquired on 08 January 2018 (Path:34, Frame: 478) as master image and 14 January 2018 (Path: 34, Frame: 478) as slave image with normal baseline of 77.057 m and incidence angle of 34.307 degrees were used for velocity estimation provided a temporal baseline of 6 days, making it well-suited for detecting glacier dynamics in the study region. VV polarization was used throughout all processes due to its high sensitivity. We tried using different temporal baselines and noticed that pairs with a higher temporal baseline showed lower coherence making it unsuitable for DInSAR processing.

This study utilized Ground Range Detected (GRD) products from Sentinel-1 SAR data in Interferometric Wide (IW) swath mode. The data corresponds to an ascending path (path 27, frame 107) and was acquired on September 5, 2018, and September 17, 2018. The SAR imagery is sourced from Sentinel-1 A, launched in April 2014. This study employs the C-band with a wavelength of 5.6 cm and features a 12-day repeat cycle. The data used in this study has single VV polarization (vertical transmit and vertical receive) and a nominal resolution of 5×20 m in range and azimuth directions. To enhance cross-coregistration of the two Sentinel-1 images, the SRTM 3-Second Digital Elevation Model (DEM) with 30 m resolution was utilized.

This study also utilized Landsat 8 imagery, which has a 16-day temporal resolution and captures Earth's surface at 30 m spatial resolution, with a higher 15 m resolution for its panchromatic band. Landsat 8, equipped with the Operational Land Imager (OLI) and Thermal Infrared Sensor (TIRS), features a 12-bit radiometric resolution for multispectral bands. During the ablation period (August-September), surface ice velocity was analysed using the Panchromatic band. Glacier boundaries were derived from the Randolph Glacier Inventory (RGI) version 6.0. Landsat offers advantages such as a larger footprint, shorter revisit times, systematic data acquisition, and the longest near-continuous Earth surface record, making it ideal for long-term climate studies.

Methodology

Mapping glacier surface ice velocities with SAR interferometry and SAR offset tracking

This study explores the use of DInSAR to measure glacier velocity by analyzing phase differences between pairs of SAR images acquired over time. Sentinel-1 A/B imagery, for January month is determined from the Alaska Satellite Facility's (ASF) Earth Data Login platform with 6-day temporal base line was utilized, a low baseline in

DInSAR helps maintain high coherence between SAR acquisitions, minimizing geometric distortions for more accurate results. The dataset, featuring interferometric wide-swath mode images, was selected for the study area to capture temporal variations in glacier movement.

The initial step involved co-registering the two images to ensure accurate alignment, followed by resampling them to the same dimensions using the SRTM DEM (30 m) for detailed analysis. Interferograms were generated by subtracting the phase of the master image from that of the slave image. The phase difference between two radar scenes (Slave and master) in Synthetic Aperture.

Radar (SAR) interferometry is described in Eq. (1).

$$\Phi_{\text{Interferogram}} = \Phi_{\text{Displacement}} + \Phi_{\text{Topography}} + \Phi_{\text{Atm}} + \Phi_{\text{Noise}} \quad (1)$$

Where $\Phi_{\text{Interferogram}}$ = Interferometric phase,

$\Phi_{\text{Displacement}}$ = Phase difference due to surface displacement (e.g., deformation or motion),

$\Phi_{\text{Topography}}$ = Phase difference due to topography,

Φ_{Atm} = Phase difference caused by atmospheric conditions,

Φ_{Noise} = Phase noise due to sensor limitations or random errors.

Coherence maps value ranges b/w the 0 to 1 were also created to evaluate the interferogram's quality, ensuring reliable phase data as the applicability of DInSAR greatly depends on the degree of coherence between the image pair. To enhance the interferogram's clarity and retain deformation fringes, filtering was applied using the Goldstein filter. Since the interferogram contains wrapped phase values, phase unwrapping was conducted with the Minimum Cost Flow (MCF) algorithm and by applying the coherence thresholds of 0.25, for transforming the cyclic phase into a continuous surface. The unwrapped phase is converted into displacement by using the Eq. (2).

$$\Delta R = (-\lambda/4\pi) * \Delta\Phi \quad (2)$$

Where, ΔR : Displacement in LOS direction.

$\Delta\Phi$: Unwrapped phase.

λ : Wavelength.

The displacement was divided by the time gap between the acquisitions of the master and slave images to calculate the Line of Sight (LOS) velocity. Displacement measured towards a satellite offers limited applicability in glaciological studies. The displacement is therefore decomposed in the direction of the surface slope for more insightful analysis, and it is frequently believed that glacier flow is parallel to the surface. These LOS velocities were further converted into the glacier surface velocity. It can be decomposed using Eq. (3)⁵⁶.

$$V_{(glac)} = \frac{U_w}{\cos(\alpha)_* \cos(\Phi)_* \sin(\theta) + \cos(\theta)_* \cos(\alpha)} \quad (3)$$

The Equation depicts the LOS velocity (U_w) in glacier flow direction (V_{glac}) using slope (α), aspect angle with respect to radar direction (Φ) and incidence angle (θ). The entire processing for DInSAR was carried out using ENVI SARscape 5.6. This software specializes in advanced SAR data analysis. It was utilized to ensure accurate and efficient DInSAR processing. Additionally, ArcGIS 10.8 was used for map composition to effectively visualize the results.

The SAR offset tracking methodology employed selected Sentinel-1 GRD IW product image pairs with low temporal baselines and precise orbit files for orbit corrections. Data processing of Sentinel-1 radar images was performed using the specialized software SNAP (Sentinel Application Platform). The offset tracking method was applied to a sequence of acquisitions 12 days apart.

The normalized cross-correlation coefficient is evaluated by applying the following Eq. (4)

$$\rho(x, y) = \frac{\sum x, y (f(x, y) - f') (t(x - u, y - v) - t')}{\sqrt{\sum x, y (f(x, y) - f')^2 \sum x, y (t(x - u, y - v) - t')^2}} \quad (4)$$

Where, f' and t' are the mean of intensities in $f(x, y)$ and $t(x - u, y - v)$ respectively. The value of the coefficient lies in the range of -1 to 1 .

Various window sizes were tested, and a 64×64 window with a pixel spacing of 20×20 pixels yielded optimal results, this research used high-resolution STRM DEM for image pair coregistration, reducing errors in the calculated glacier velocity. Movement between the slave and master pictures is shown by the cross-correlation surface's peak property (Eq. 5)⁵⁷.

$$D = \sqrt{(R_{\text{range}} \Delta x_{\text{range}})^2 + (R_{\text{azi}} \Delta x_{\text{azi}})^2} \quad (5)$$

Where R_{range} , R_{azi} are pixel spacing in range and azimuth directions, and Δx_{range} , Δx_{azi} are pixel shifts in range and azimuth direction.

Mapping glacier surface ice velocity with optical data

The advanced sub-pixel co-registration technique, COSI-Corr, developed by Leprince et al.⁵⁸, as an ENVI add-on, facilitates precise glacier dynamics analysis. The study by Scherler et al.⁵⁹, presents a methodology for studying glacier dynamics. This method addresses inaccuracies in DEMs and satellite scanning errors, enhancing co-

registration precision. Preprocessing involves orthorectifying raw satellite images using adjusted ground control points (GCPs), creating a reference image for aligning subsequent images. Two time-series images are then correlated at the sub-pixel level using a frequency correlator, yielding horizontal displacements (east-west and north-south) and a signal-to-noise ratio (SNR) for reliability assessment. The results are refined through three filters: an SNR filter to remove poorly correlated pixels, a direction filter to align data with expected glacier flow, and a magnitude filter to account for gradual movement changes. The final displacements are standardized annually for consistent temporal comparison, allowing detailed insights into glacier dynamics.

Data processing for ERT data

Electrical Resistivity Tomography (ERT) surveys were conducted along selected profiles on the lateral and terminal moraines of the Durung Drung glacier using the Syscal-Pro Switch-72 system.

A combination of Dipole–Dipole, Wenner, and Wenner–Schlumberger arrays was employed to balance resolution and depth of investigation. The Dipole–Dipole array was selected for its higher sensitivity to lateral resistivity variations and structural features, whereas the Wenner and Wenner–Schlumberger arrays provided improved signal strength and better vertical resolution. An electrode spacing of 5 m was adopted as a compromise between depth penetration (up to ~45 m) and resolution of near-surface heterogeneities, which are critical in moraine environments. The acquired apparent resistivity data were processed using RES2DINV software⁶⁰. Prior to inversion, data quality was assessed and noisy data points associated with poor electrode contact or high contact resistance were removed. Topographic corrections were incorporated using field-measured elevation data to improve model accuracy.

The inversion was carried out using a smoothness-constrained least-squares approach based on a quasi-Newton (Gauss–Newton) optimization scheme. An L2-norm (least-squares) inversion was applied to obtain smoothly varying resistivity models, which are appropriate for representing gradual transitions between unfrozen, frozen, and lithological units in glacial environments. The initial model was defined as a homogeneous half-space with resistivity equal to the mean of the measured apparent resistivity values. Model discretization was automatically generated by RES2DINV, with block sizes adjusted according to electrode spacing and array configuration. Sensitivity to side effects and boundary conditions was minimized by extending the model domain within the electrode spread. Multiple iterations were performed until convergence was achieved between measured and calculated apparent resistivity values, as indicated by the reduction in RMS error. The final inversion results were used to generate 2D resistivity tomograms, which were interpreted in terms of subsurface lithology, frozen sediments, and hydrological conditions within the lateral and terminal moraines. Three electrode configurations (Dipole–Dipole, Wenner, and Wenner–Schlumberger) were used to improve the reliability of subsurface imaging in the moraine environment. Each array provides different sensitivity to lateral and vertical resistivity variations, allowing cross-validation of the inversion results. The datasets were processed separately to preserve array-specific resolution characteristics and to assess the consistency of subsurface resistivity patterns, thereby reducing interpretation uncertainty. The maximum depth of investigation was estimated from the RES2DINV inversion output based on electrode spacing, array configuration, and sensitivity distribution of the model blocks. These depths represent approximate effective investigation depths rather than exact penetration limits, and no separate depth-of-investigation (DOI) or sensitivity analysis was performed.

Results and discussion

ERT profiles and interpretation

In Fig. 3 (Dipole–Dipole array), the lateral moraine profiles extend up to 180 m and 360 m, with RMS errors of 4.0% and 6.1%, respectively. The resistivity sections show clear spatial variability, reflecting a heterogeneous subsurface typical of moraine environments.

The upper layer is characterized by relatively low resistivity values, which are interpreted as unconsolidated, water-saturated debris. At greater depths, moderate to high resistivity zones are observed, which may correspond to more compacted sediments, frozen ground, or bedrock. However, it should be noted that resistivity values alone cannot uniquely distinguish specific lithologies, as they are influenced by multiple factors such as moisture content, porosity, temperature, and degree of freezing.

Several localized resistivity contrasts are visible along the profiles. These variations may indicate changes in material properties, such as transitions between coarse debris, frozen sediments, and more consolidated units. However, these features are interpreted as zones of heterogeneity rather than definitive structural discontinuities. No direct evidence of faulting can be conclusively inferred from the ERT data alone.

Profiles 3(a) and 3(b) display broadly similar resistivity patterns, suggesting lateral continuity of subsurface conditions at the scale of investigation. The observed variability highlights the complex internal structure of the moraine, but interpretations remain non-unique in the absence of borehole or ground-truth data.

Wenner arrays with 5 m spacing (Lateral Moraine)

In Fig. 4 (Wenner array), the lateral moraine profiles extend approximately 180 m and 360 m, with RMS errors of 1.31% and 2.2%, respectively. The resistivity sections show clear spatial variability, reflecting heterogeneous subsurface conditions typical of moraine environments.

The upper part of the profiles is characterized by relatively low resistivity values, which likely represent unconsolidated and water-saturated debris material. At intermediate depths, moderately higher resistivity zones are observed, which may correspond to compacted sediments, coarse debris, or partially frozen ground. The deeper sections show comparatively higher resistivity values that could indicate more consolidated materials or bedrock units; however, resistivity data alone do not allow precise identification of specific lithologies.

Resistivity contrasts observed along the profiles indicate variations in subsurface material properties and moisture conditions rather than definitive structural features. Similar resistivity variations are visible at several

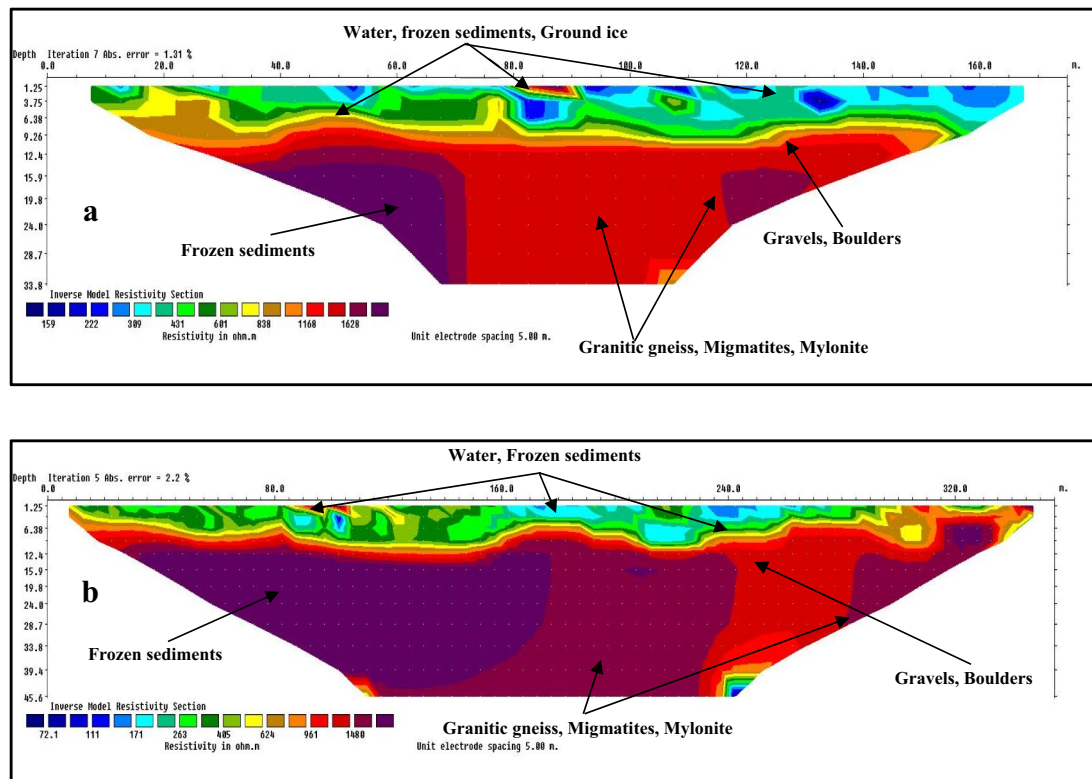


Fig. 4. Resistivity sections obtained after inversion of ERT lines (a) and (b) carried out along the Lateral Moraine of the Durung Drung glacier (Wenner).

locations along the profiles, suggesting general heterogeneity within the moraine. Profiles 4(a) and 4(b) show broadly consistent resistivity patterns, indicating lateral continuity of subsurface conditions.

Overall, the Wenner array results highlight the complex internal structure of the lateral moraine, while interpretations remain non-unique in the absence of borehole or direct geological validation.

Wenner-Schlumberger arrays with 5 m spacing (Lateral Moraine)

In Fig. 5 (Wenner–Schlumberger array), the lateral moraine extends approximately 180 m and 360 m, with RMS errors of 3.4% and 3.5%, respectively. The resistivity patterns are generally consistent with those observed in the Wenner array, indicating comparable subsurface conditions.

Lower resistivity values in the upper layer likely represent unconsolidated and water-rich debris material. Intermediate resistivity zones may indicate coarse sediments or partially frozen ground, while deeper high-resistivity regions could correspond to more consolidated material or bedrock. However, it is important to note that resistivity values are influenced by several factors, including moisture content, porosity, temperature, and degree of freezing, making precise lithological identification uncertain.

Variations in resistivity along the profiles reflect changes in subsurface properties and moraine composition rather than clear structural discontinuities. The similarity between profiles 5(a) and 5(b) suggests consistent subsurface conditions across the lateral moraine.

These results support the interpretation of a heterogeneous moraine structure with variable moisture and frozen sediment distribution, while acknowledging the limitations of geophysical data without ground-truth validation.

In Fig. 6 (Dipole–Dipole array), the terminal moraine extends approximately 180 m and 360 m, with RMS errors of 8.8% and 6.1%, respectively. The resistivity sections show clear lateral and vertical variations, indicating heterogeneous subsurface conditions.

The upper layer is characterized by relatively low resistivity values, which likely correspond to unconsolidated debris and moraine material. At greater depths, moderate to high resistivity zones are observed, which may represent compacted sediments or partially frozen ground. The deepest regions show higher resistivity values that could indicate more consolidated material or bedrock; however, these interpretations remain inferential.

Resistivity contrasts observed along the profiles represent changes in subsurface material properties and moisture conditions rather than definitive fault structures. Similar resistivity variations are present at multiple locations, suggesting general heterogeneity within the terminal moraine. Profiles 6(a) and 6(b) display comparable resistivity trends, indicating lateral continuity of subsurface conditions.

Overall, the Dipole–Dipole results provide insight into the internal variability of the terminal moraine while emphasizing the non-unique nature of geophysical interpretations.

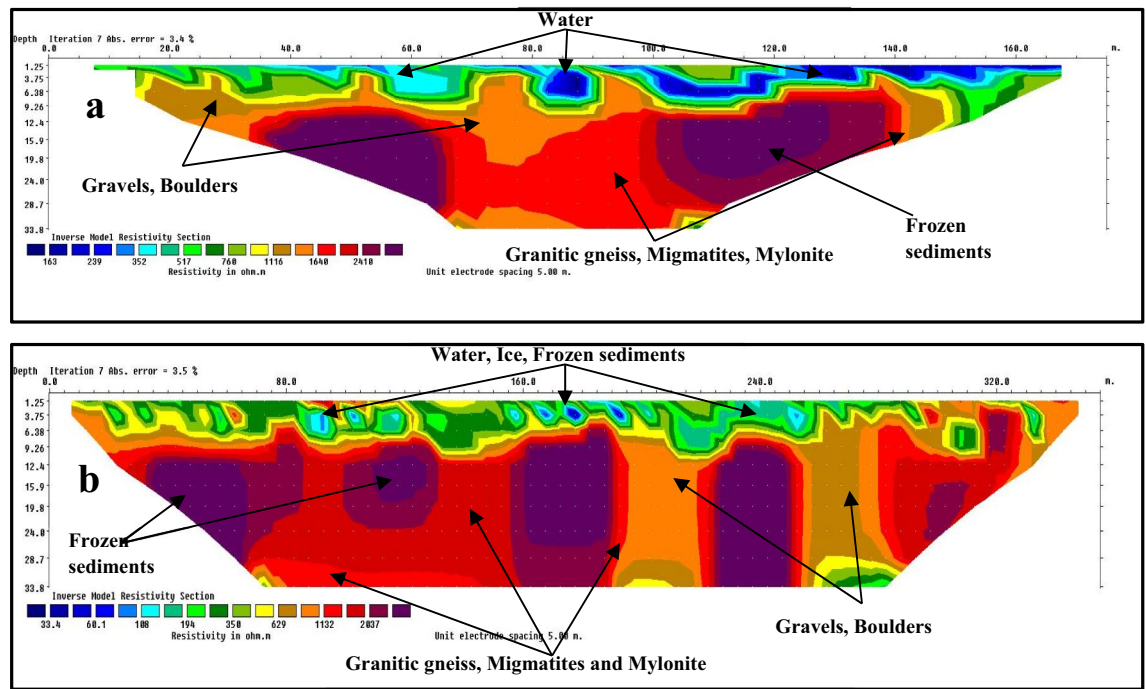


Fig. 5. Resistivity sections obtained after inversion of ERT lines (a) and (b) carried out along the Lateral Moraine of the Durung Drung glacier (Wenner-Schlumberger).

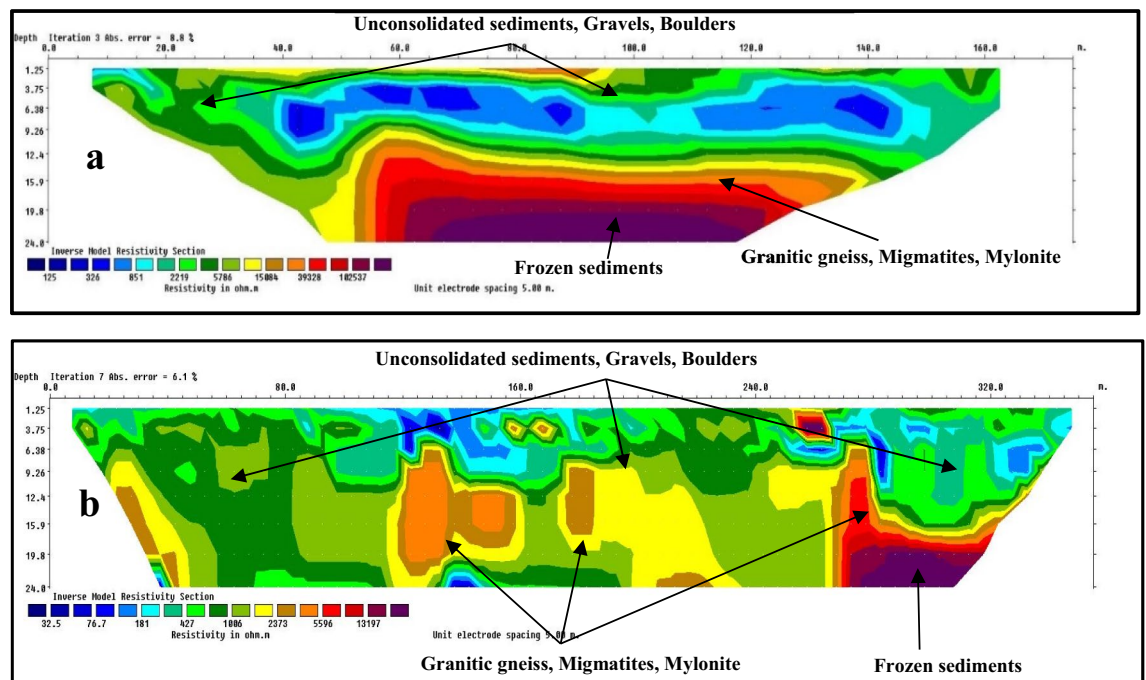


Fig. 6. Resistivity sections obtained after inversion of ERT lines (a) and (b) carried out along the Terminal Moraine of the Durung Drung glacier (Dipole-Dipole).

Wenner arrays with 5 m spacing (Terminal Moraine)

In Fig. 7 (Wenner array), the terminal moraine extends approximately 180 m and 360 m, with RMS errors of 13.4% and 3.9%, respectively. The resistivity sections reveal distinct spatial variations typical of moraine environments.

The upper portions of the profiles show relatively low resistivity values, likely indicating unconsolidated and water-bearing debris material. Intermediate resistivity zones may correspond to coarse moraine sediments or partially frozen ground, while deeper high-resistivity regions could indicate more consolidated material or bedrock.

The observed resistivity contrasts reflect variations in subsurface composition and moisture distribution rather than clear structural discontinuities. Similar patterns occur at multiple locations along the profiles, suggesting general heterogeneity rather than discrete fault zones. Profiles 7(a) and 7(b) show broadly consistent resistivity patterns, supporting lateral continuity within the terminal moraine.

These findings highlight the complex internal structure of the terminal moraine and the variability of subsurface materials under glacial conditions.

Wenner-Schlumberger arrays with 5 m spacing (Terminal Moraine)

In Fig. 8 (Wenner-Schlumberger array), the terminal moraine extends approximately 360 m, with RMS errors of 2.4% and 19.9% for profiles 8(a) and 8(b), respectively. The resistivity sections show clear spatial variability, reflecting heterogeneous subsurface conditions.

Low resistivity values in the upper layer likely represent unconsolidated debris and water-rich moraine material, while intermediate resistivity zones may indicate compacted sediments or partially frozen ground. Higher resistivity values at greater depths could correspond to more consolidated material or bedrock; however, resistivity data alone cannot uniquely identify specific lithologies. The relatively high RMS misfit (19.9%) in profile 8(b) is likely related to heterogeneous moraine material, variable moisture conditions, and possible electrode contact issues, which introduce additional uncertainty in the inversion results. Therefore, interpretations for this profile should be considered with caution. Overall, both profiles show broadly similar resistivity trends, suggesting general continuity of subsurface conditions, while highlighting the heterogeneous nature of the terminal moraine.

The different resistivity layer architecture identified in the lateral and terminal moraines has important implications for understanding the geomorphic evolution and present thermal regime of the Durung Drung Glacier (Table 1). The low-resistivity surface reflects water-saturated unconsolidated sediments, indicating active meltwater infiltration and ongoing reworking of moraine surfaces^{61,62}. The presence of frozen sediments rather than massive glacier ice in the intermediate and deeper layers reveals that the moraines are no longer ice-cored; instead, they contain refrozen permafrost-like materials formed after the melt-out of earlier englacial or buried ice^{63,64}. Such transitions from ice-cored to ice-free but partially frozen moraines are consistent with glaciers undergoing sustained thinning and recession across the Himalaya and other high-mountain regions^{65,66}.

The deeper frozen lenses in the terminal moraine, compared with mid-horizon frozen sediments in the lateral moraine, suggest differential insulation, sediment thickness, and post-depositional cooling patterns also observed in recent Himalayan ERT-based moraine investigations^{67,68}. The repeated low-resistivity passages marking structural discontinuities further highlight preferential groundwater pathways, which influence internal thaw processes, hydrological routing, and moraine stability^{11,69}. Such internal heterogeneities resemble those documented in actively degrading moraine and permafrost bodies worldwide, emphasizing the sensitivity of unfrozen–frozen transitions to changing climatic conditions⁶⁴.

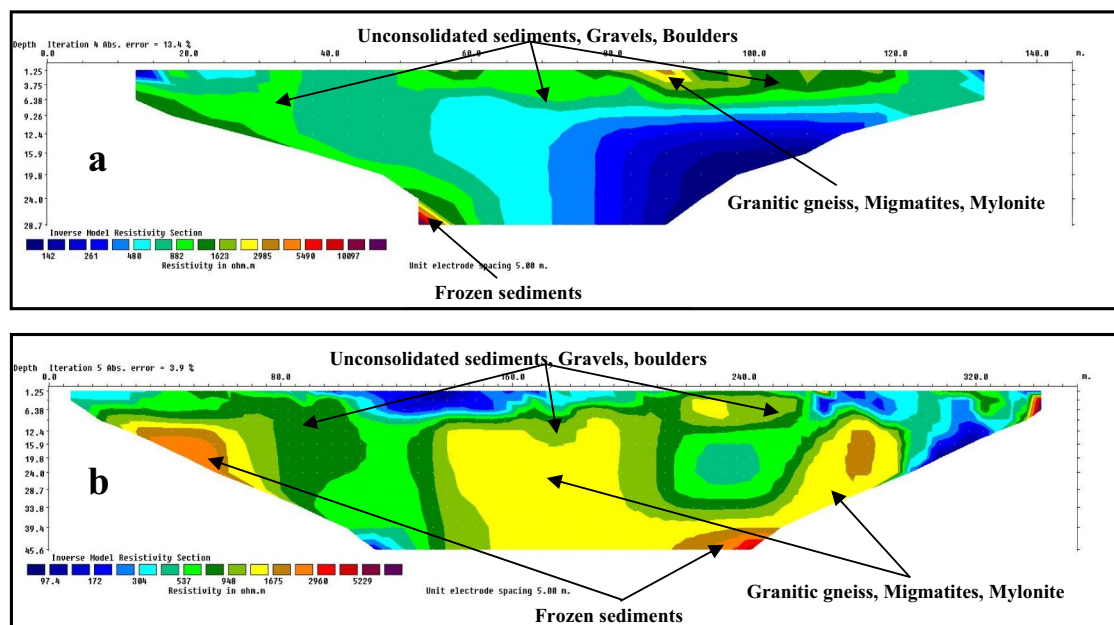


Fig. 7. Resistivity sections obtained after inversion of ERT lines (a) and (b) carried out along the Terminal Moraine of the Durung Drung glacier (Wenner).

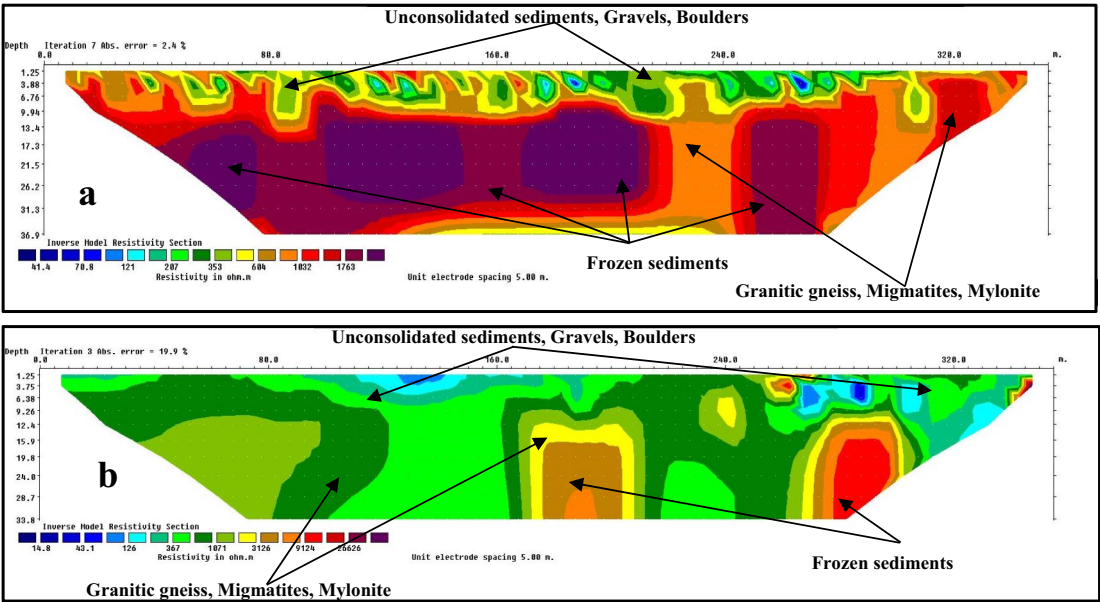


Fig. 8. Resistivity sections obtained after inversion of ERT lines (a) and (b) carried out along the Terminal Moraine of the Durung Drung glacier (Wenner-Schlumberger).

Location of the Profiles	Array Types	Maximum depth reached (m)	Average Resistivity Range (Ω-m)	Interpreted Units/Layers
Lateral Moraine	Dipole-Dipole	25.2 m	< 88.6 to > 3832	Upper: Water/debris sediments Middle: Frozen sediments, gravels, boulders Lower: Granitic gneiss, migmatites, mylonite
	Wenner	45.6 m		
	Wenner-Schlumberger	33.8 m		
Terminal Moraine	Dipole-Dipole	24.0 m	< 100 to > 5000	Upper: Unconsolidated sediments, gravels, boulders Middle: Granitic gneiss, migmatites, mylonite Lower: Frozen sediments
	Wenner	45.6 m		
	Wenner-Schlumberger	36.9 m		

Table 1. Summary of ERT results for lateral and terminal moraines of Durung Drung glacier.

Collectively, these resistivity patterns indicate advanced deglaciation, evolving discontinuous permafrost conditions, and a shift toward a hydrologically and mechanically sensitive moraine system strong evidence of ongoing climatic forcing and glacier retreat in the Durung Drung catchment^{70,71}. The ongoing frontal recession has also facilitated the progressive enlargement of the moraine-dammed lake at the glacier snout due to continued ice loss and increases meltwater accumulation. The weakening of internal thaw features identified in the moraine further promote seepage, sediment remobilization, and basin overdeepening, collectively supporting the rapid expansion of the proglacial lake in recent years.

Surface ice velocity estimation

The surface ice velocity was measured using the DInSAR, offset tracking and optical data feature tracking techniques. The results obtained through the DInSAR technique are depicted in Figs. 9, 10, 11 and 12. Figure 9 shows the glacier coherence map for the Durung Drung glacier as the coherence along the main parts and central flow line of the glacier is more than 0.8 due to the 6 days between both master (8 January 2018) and slave image (14 January 2018) and the Fig. 10 shows the interferogram after applying the Goldstein filter. The glacier LoS velocity for the year 2018 January month which ranges from +0.12 m d⁻¹ to -0.33 m d⁻¹, here the negative values indicate the values away from the satellite and positive value indicates the velocity values towards the satellite look direction (Fig. 11).

The surface ice velocity of the Durung Drung glacier was determined using DInSAR and SAR offset tracking measurements which were validated against velocity estimations from optical data. The glacier surface ice velocity using DInSAR measurements is shown in Fig. 12 and its values range from 0.19 m a⁻¹ to 120.3 m a⁻¹. The glacier's greatest surface velocity in the ablation zone was 120.3 m a⁻¹ (0.32 m d⁻¹). Figure 12 (b) and (c) shows the transect analysis over different zones of the glacier. The A-to-B profile depicts glacier velocity from its snout to the accumulation region covering a total distance of 23 km, measured in meters per year (m a⁻¹). Velocity starts low near the snout from 0.19 m a⁻¹, rises significantly between 8,000 and 12,000 m to 100 m a⁻¹ along the central flow line, and fluctuates further due to changes in terrain and ice dynamics. Peaks exceeding 90

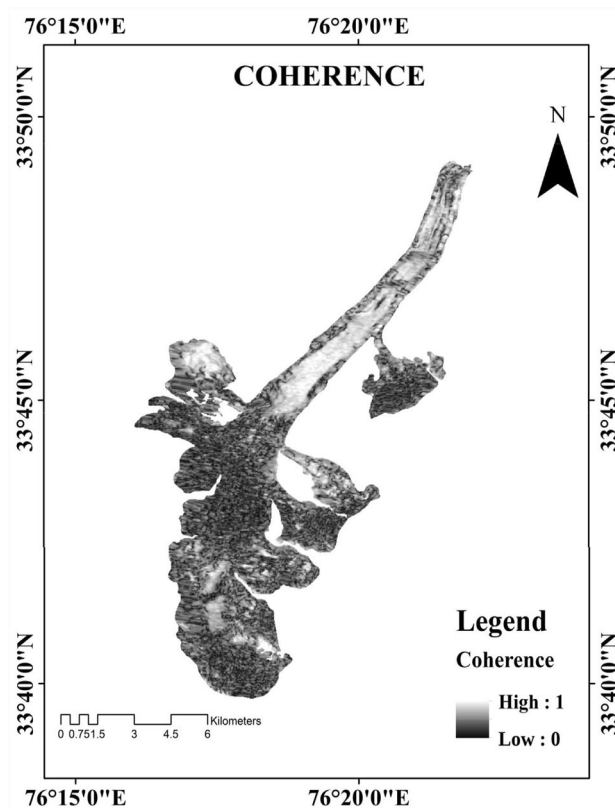


Fig. 9. Map showing the Coherence of Durung Drung glacier.

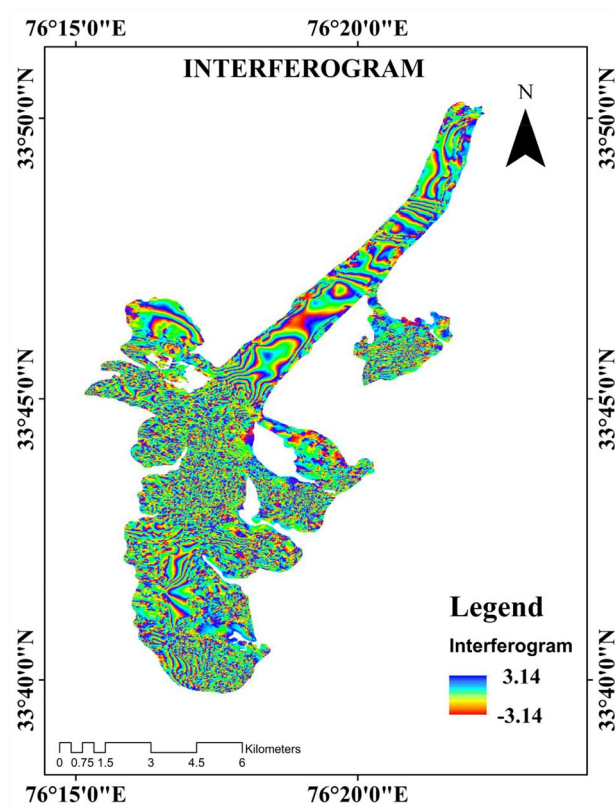


Fig. 10. Map showing the interferogram after applying the Goldstein filter.

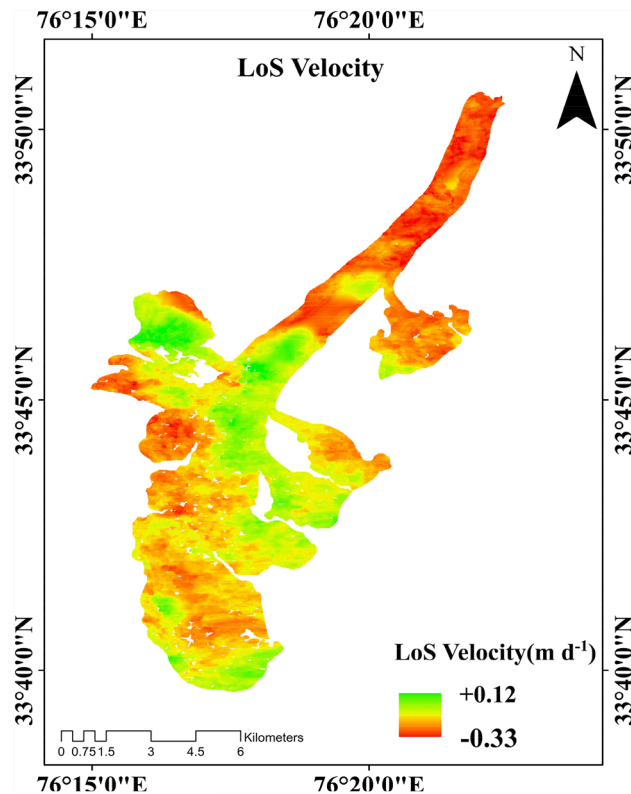


Fig. 11. Map showing the LoS velocity in January 2018.

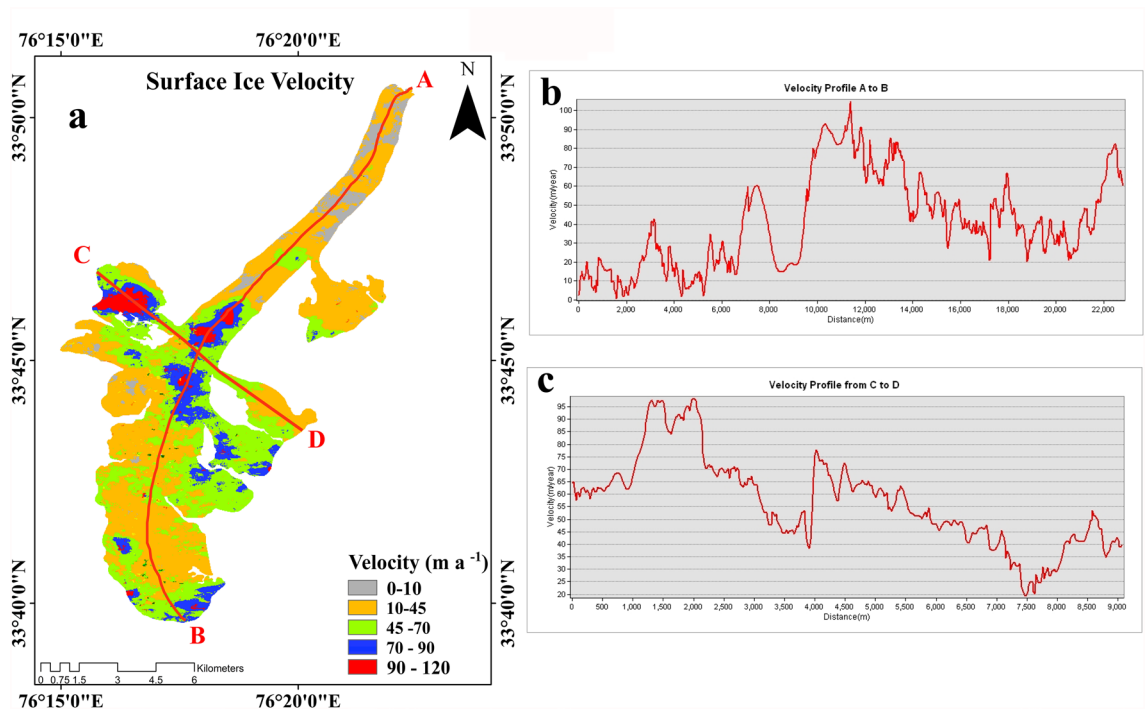


Fig. 12. (a) Map showing the glacier surface ice velocity for the year 2017-18 calculated using the sentinel 1 A/B data (b) and (c): Transect analysis using longitudinal and cross-section profiles respectively.

m a^{-1} indicate steep slopes or thick ice, reflecting the influence of slope, ice thickness, and subglacial conditions. The C-to-D profile illustrates glacier velocity from left to the right side of the glacier. Velocity peaks at $\sim 95 \text{ m a}^{-1}$ between 1,000 and 2,000 m distance and gradually decreases with fluctuations. A notable drop near 7,500 m highlights the impact of slope and ice conditions on glacier flow. When top meltwater approaches the bottom in the summer, it also functions as a fluid and reduces the contact between bedrock and glacier ice⁷². As a result, the glaciers move more quickly. The DInSAR ice velocities do not account for such annual velocity variations. These seasonal fluctuations may be the cause of the variations in the velocity curve⁷³.

The SAR offset tracking velocity ranges from a minimum of 0.0009 m a^{-1} to a maximum of 139 m a^{-1} as shown in Fig. 13, with the highest velocities observed along the central flowline in the ablation zone.

Optical data measurements reveal that the surface ice velocity is significantly higher in the ablation zone compared to the accumulation zone, while the areas near the snout exhibit slower movement. The low velocity in the accumulation zone can be linked to the degraded condition of the snow cover. The glacier's surface ice velocities for 2017-18 is 138 m a^{-1} as illustrated in Fig. 14, showing a pulsating pattern that may result from hydro-meteorological factors such as variations in mass balance, seasonal snowfall, and the presence of crevasses. To address data voids, interpolation techniques were applied to fill gaps during the correlation process.

The glacier velocity estimates derived from DInSAR, offset tracking, and cross-correlation techniques exhibit minor variations yet remain broadly consistent, reflecting the reliability of the integrated approach. The maximum velocity derived from DInSAR is approximately 120 m a^{-1} , whereas the offset tracking and cross-correlation methods yield values of around 139 m a^{-1} and 138 m a^{-1} , respectively. The slight differences primarily stem from the nature of the datasets and processing methodologies employed. The DInSAR technique, which depends on phase differences between SAR acquisitions, utilized January (peak winter) data to ensure high temporal coherence. During winter, the glacier surface remains predominantly frozen and stable, minimizing melting and surface changes. These stable conditions effectively reduce temporal decorrelation and phase noise, allowing the interferometric phase to more accurately represent true surface displacement rather than atmospheric or surface-change artifacts. Consequently, winter-based interferograms produce cleaner phase signals and yield more reliable displacement estimates. In contrast, offset tracking and cross-correlation both pixel-tracking methods independent of coherence were applied to ablation-season data, capturing the cumulative annual displacement. These techniques are less affected by temporal decorrelation and are well suited for larger displacement magnitudes, hence complementing the DInSAR-derived measurements.

The unusually high glacier velocities in the middle part of the glacier are mainly due to stronger ice deformation and basal sliding. This central zone represents the main glacier trunk, where both ice thickness ($> 400 \text{ m}$), and surface slope are greater, creating more stress that speeds up ice movement⁷⁴. In addition, tributary glaciers joining the main glacier add more ice, increasing the flow rate. During warmer periods, meltwater may reach the glacier bed, reduce friction and allow the ice to slide more easily. Altogether, these factors steeper slope,

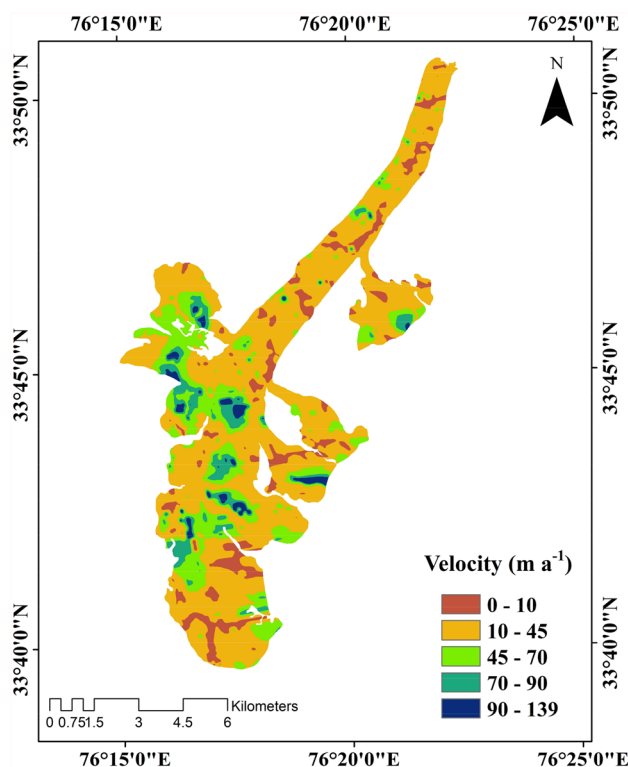


Fig. 13. Map showing the glacier surface ice velocity for the year 2017-18 calculated using the Sentinel 1 A SAR offset tracking technique.

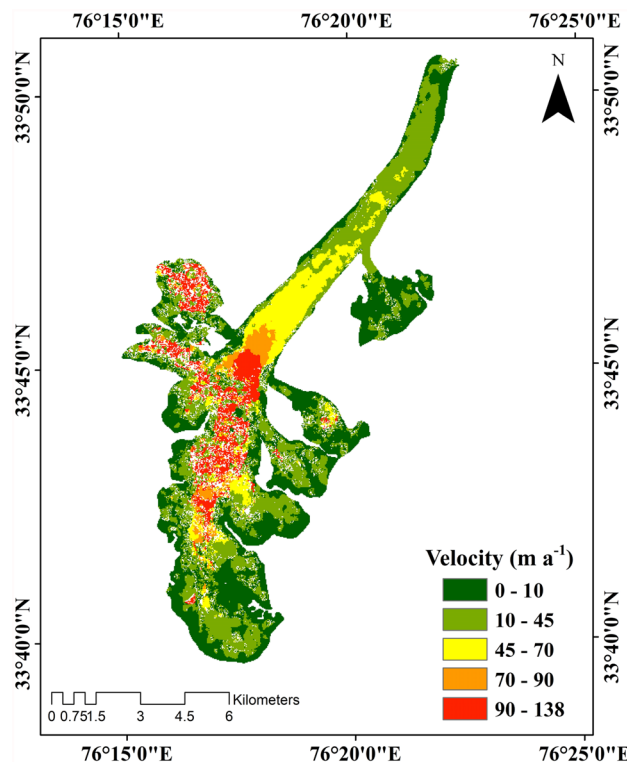


Fig. 14. Surface Ice Velocity calculation on Durung Drung glacier using the sub-pixel image correlation technique for the year 2017–18.

thicker ice, and meltwater at the base cause higher velocities in the middle section, which is typical for large valley glaciers in high mountain regions.

Overall, the velocity results from all three methods demonstrate consistent spatial trends across the Durung Drung Glacier. DInSAR and SAR offset tracking both reveal maximum velocities concentrated within the ablation zone, with offset tracking capturing a broader velocity spectrum from low to high values. Similarly, optical data highlight elevated ice motion in the ablation area and slower flow near the snout and accumulation zone, consistent with the DInSAR observations of reduced velocities in the upper glacier region. Although the absolute values differ slightly due to methodological and temporal variations, all three techniques converge in depicting the glacier's dynamic behavior, thereby reinforcing the robustness and complementarity of the multi-sensor velocity assessment.

Area changes

From 1976 to 2020, the Durung Drung glacier depicts a significant reduction in area, losing 8.16 km², equivalent to a decrease of 10.73%. Significant fluctuations were observed on a decadal scale. The Durung Drung Glacier has undergone a frontal retreat of approximately 812.36 m between 1976 and 2020, reflecting the combined effects of climatic and geomorphological drivers. The retreat coincides with a regional warming trend of about 0.5 °C per decade and a decline in winter snowfall by nearly 10–15%, which collectively enhance surface melting and limit accumulation in the accumulation zone^{75,76}. Similar magnitudes of recession have been recorded for glaciers in the Zaskar, Suru, and Ladakh regions, showing retreat distances between 700 m and 1.2 km and average mass loss rates of -0.35 ± 0.12 m w.e. a⁻¹ over the past four decades^{1,77} (Fig. 15). The observed glacier dynamics exhibit a clear relationship with both climatic variability and geomorphological factors within the Durung Drung Glacier system. The measured surface velocities and snout retreat correspond to regional warming trends, which enhance surface melting and promote basal lubrication, thereby accelerating ice flow. However, the spatial variability in velocity and resistivity patterns suggests that local geomorphological controls modulate this climatic influence. Areas with thick supraglacial debris exhibit reduced melt rates and consequently lower flow velocities, whereas clean-ice zones and steep slopes facilitate faster ice motion. The ERT-derived subsurface heterogeneity, including variations in debris, frozen sediments, and ground ice, further indicates differential thermal and hydrological regimes influencing basal motion. These findings highlight that the glacier's response is governed by the combined effects of temperature-driven melting, subglacial hydrology, and terrain-induced mechanical resistance, underscoring the importance of integrating geophysical and satellite-based observations to understand glacier–climate–geomorphology interactions in the Durung Drung glacier^{75,77}. The highest retreat occurred between 2010 and 2015, measuring 325.39 m, while the lowest retreat was 65 m between 2015 and 2020⁷¹. To validate the satellite results, fieldwork was done between June and October from 2019 to 2020. Exact measurements were made both horizontally and vertically using the Leica DGPS CS10/15 system, which is renowned for its great precision in mountainous areas. The study involved measuring the morphology of glacier,

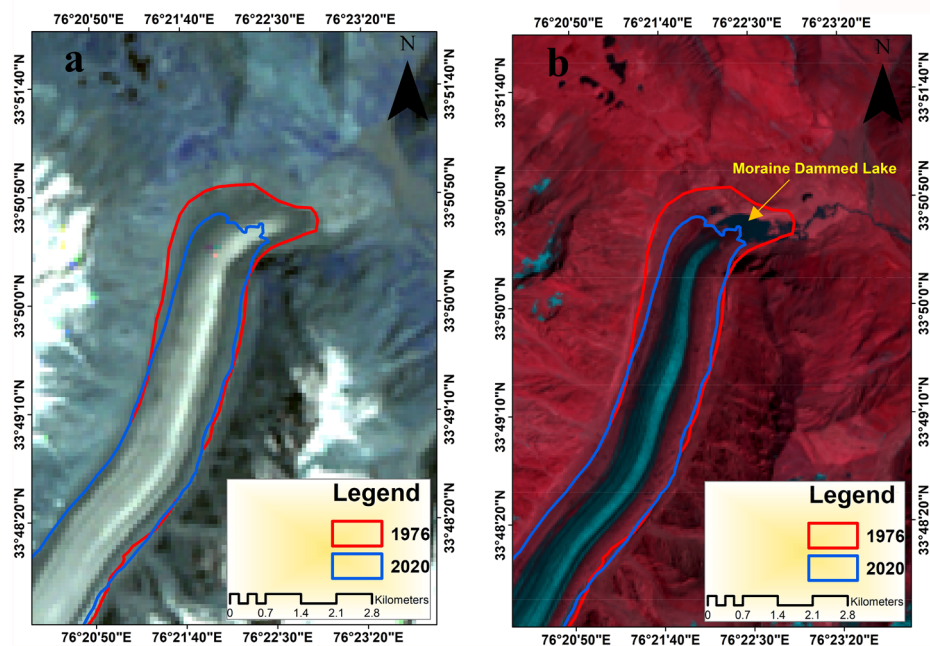


Fig. 15. The fluctuation pattern of the glacier terminus is illustrated using: (a) a 1976 Landsat 2 Multispectral Scanner image and (b) a Landsat 8 OLI satellite image, highlighting the newly formed moraine-dammed lake at the Durung Drung Glacier.

including the exact locations of debris cover. According to the DGPS study, a notable snout retreat of 14.93 ± 0.5 m, 16.62 ± 0.5 m, & 21.43 ± 0.5 m was found over the 2018–2019, 2019–2020, & 2020–2022 period respectively in the Durung Drung glacier's frontal zone. In light of continuous rising temperatures, these combined approaches improve the accuracy of evaluations of glacial change and help to produce more accurate forecasts in future.

The complicated climate and terrain of the Himalayan region, along with the scarcity of meteorological data, make it complex to understand the energy budgets of the glaciers there⁷⁵. Surficial debris monitoring is extremely important since it has a significant impact on glacier dynamics and ablation processes. The Himalayas are dense with debris-covered glaciers, and existence of supraglacial debris above the glacier surface drastically changes the reaction of glaciers to changing climate⁷⁸. According to the proportion of debris-covered the whole glacier surface, clean glaciers have the largest mass loss, followed by glaciers with sporadic debris coverage and glaciers with heavy debris coverage, which have the lowest mass loss⁷⁹. Considering the continual existence of geomorphic structures like ice cliffs and glacial lakes certain debris-covered glaciers might encounter mass loss comparable to those of pristine glaciers⁸⁰. Even though these features are specific, greatly influence the glacier base's retreat⁸¹. To understand the geomorphic features' link with the glacier mass and related volumetric shifts it is crucial to evaluate them. 5.28% of the Durung Drung glacier area is overlaid by debris, according to projections of debris coverage derived from the 2013 OLI data³⁶. Different topographic contrasts or variations in the adjacent lithology could be the cause of the fluctuating debris cover of the Durung Drung Glacier²⁷.

The Durung Drung glacier sculpted the Higher or Greater Himalayan crystalline rocks, which are mostly composed of schists, phyllites, gneisses, and migmatites⁸². The proglacial lake at the glacier front began to form in the early 1990s, although significant expansion did not occur until the late 1990s and early 2000s. Initially, only a few small ponds were present in 1992; however, between 2000 and 2009, these increased both in number and size. By 2011–2013, the individual ponds had coalesced to form larger lakes, which became established at the northern tip of the glacier snout³⁶. Lakes cause the phenomenon of ice-shattering to accelerate and disintegrate the terminal. They also transfer heat to the ice forefront, accelerating melting and aiding retreat⁸³. High-resolution imagery shows that ice-shattering has begun in this lake as well, which could be a major factor in the current spike in glacier retreat patterns³⁶. The extent of the debris may potentially have an impact on glacier ablation⁸⁴. Glacier dynamics are influenced by supraglacial debris-covered. Debris cover has the potential to accelerate or slow down the ablation process, based on its thickness²⁷. Debris in Durung Drung glacier reached 5.5% (3.8 km^2) of the total area. Between 2015 and 2019, the bottom ablation zone of the Durung Drung glacier has been adorned with thin or dispersed debris, which accelerates the terminus retreat of glacier⁸⁵. The degree of melting glaciers fluctuates when there is more than 20% of debris cover a value of zero but it is high when there is no debris. A debris load alters a glacier's reaction to climatic change and affects the behaviour of the terminus. When there is a thin debris layer (less than 5 cm), surface ablation values tend to increase but when there is a heavy debris cover (more than 5 cm), surface ablation values are dramatically lowered²⁷. Numerous localized geographical elements, like glacier dimensions, direction, slope, hypsometry, and debris coverage, have a substantial impact on glacial activities. The Durung Drung glacier lost about 812.36 m of length and 10.73% of its total area between 1976 and 2020⁷¹. Except for alpine peri-glacial environments, the integration of DInSAR

and ERT approaches has not been employed very often till now⁸⁶. Additionally, there are very few studies that use DInSAR ground dynamics in conjunction with geophysical measurements. Villarroel et al.⁸⁷, studied the internal dynamics of the glacier of semi-arid Andes of Argentina using DInSAR and ERT approaches. A similar study was conducted by Kunz et al.,⁸⁸ in the alpine glacier to investigate the internal framework and behavior of the moraine complex using DInSAR and geophysical methods.

Limitations of the study

Despite the integrated application of ERT, SAR, and DGPS, this study has certain limitations. Subsurface interpretations are constrained by the spatially limited ERT profiles and the absence of direct ground validation such as borehole or thermal measurements, which may introduce ambiguity in distinguishing frozen sediments from ice-rich debris. Ice-flow velocities derived from SAR represent discrete temporal windows and may not fully capture seasonal or long-term variability and moraine stability is inferred indirectly rather than quantitatively modelled. Future research should focus on 3D and time-lapse geophysical surveys, extended multi-year velocity analysis, in situ thermal monitoring, and numerical modeling of moraine stability and lake evolution to strengthen hazard assessments under ongoing climatic warming.

Conclusion

This investigation demonstrates that the combined application of ERT, DGPS, and SAR-derived velocity fields provides a robust and complementary framework for resolving the coupled surface–subsurface dynamics of the Durung Drung Glacier and its periglacial environment. The resistivity stratigraphy obtained from ERT imaging reveals a systematic transition from previously ice-cored moraine structures to thermally sensitive, debris-dominated deposits hosting discontinuous ground-ice bodies. This transition reflects the progressive melt-out of buried ice and the development of shallow permafrost, leading to increased hydrological permeability and reduced internal cohesion of moraine materials, which emerge as key controls on moraine stability under ongoing climatic warming. Surface kinematic parameters derived from Sentinel-1 A/B DInSAR, SAR offset tracking, and COSI-Corr ($120\text{--}138\text{ m a}^{-1}$) confirm a dynamically active glacier responding to perturbations in surface mass balance, while the cumulative snout retreat of $52.98 \pm 0.5\text{ m}$ between 2018 and 2022 indicates sustained frontal recession and enhanced downwasting. This continued retreat, coupled with increased meltwater input, is rapidly enlarging the moraine-dammed lake due to the loss of ice support and progressive weakening of the terminal moraine, consistent with broader regional trends observed across the Zaskar Himalaya. While the integrated geophysical and remote-sensing approach provides robust insights, subsurface interpretations remain constrained by spatially limited ERT coverage and the absence of direct ground validation, and the SAR-derived velocities represent discrete temporal windows that may not fully capture seasonal or longer-term variability. Nevertheless, this study significantly improves the understanding of internal moraine architecture, identifies thaw-prone zones and structural weaknesses, and delineates evolving hydrological pathways within lateral and terminal moraine complexes, which is essential for evaluating the stability of moraine-dammed lake systems and the potential for glacial outburst-related hazards.

Data availability

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

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Additional information

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